

Under Review

Integrating Mindfulness Into The Extended Process Model Of Emotion Regulation:

The Dual-Mode Model Of Mindful Emotion Regulation

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Abstract

Extensive research has been conducted regarding how people manage their emotions. Within this research, there has been growing attention towards the role of mindfulness in emotion regulation. While prior reviews have discussed mindfulness in the context of emotion regulation, they have not provided a thorough integration using the prevailing models of emotion regulation or mindfulness. The present review discusses the Extended Process Model of Emotion Regulation and Monitoring and Acceptance Theory of mindfulness in order to propose a novel integrated framework, the Dual-mode Model of Mindful Emotion Regulation (D-MER). This model proposes two “modes” of mindfulness: Implementation and facilitation. Implementation posits that mindfulness components act as emotion regulation strategies through attentional deployment and cognitive change. Facilitation posits that mindfulness components act on emotion generation and regulation through effects on cognitive processes and positive or negative valence systems. Empirical and theoretical support for this model are discussed, specific hypotheses to guide further research are provided, and clinical implications are presented. Use of this model may identify mechanisms underlying the interaction between mindfulness and emotion regulation which can be used in ongoing affective and clinical research.

Keywords: Mindful Awareness, Acceptance, Meditation, Equanimity, Affect Regulation

Introduction

Distress and unwanted emotions are inevitable aspects of experience. As such, extensive research has been devoted to how people respond to their emotions to promote well-being and psychological health. While this large and rapidly developing body of work has made clear contributions to affective and clinical science, there remain clear gaps in the theoretical integration of key constructs—such as mindfulness. Influential models of emotion regulation and mindfulness have been presented (Baer, 2019; Grossman & Van Dam, 2011; Grossman, 2019; Gross, 2015b, 2015a; Lindsay & Creswell, 2017, 2019; Webb, Schweiger Gallo, et al., 2012) and prior work has aimed to understand mindfulness within emotion regulation (Chambers et al., 2009; Teper et al., 2013). However, prior reviews have not used prominent models to guide theoretical integration regarding the role of mindfulness in emotion regulation.

Extended Process Model of Emotion Regulation

Emotion regulation is the process by which a person engages in behaviors with the implicit or explicit goal of altering emotion frequency, intensity, duration, or type (Gross & Jazaieri, 2014). A prominent model of emotion regulation is the Extended Process Model (EPM, Gross, 2015b, 2015a; Webb, Schweiger Gallo, et al., 2012); for reviews of other models, see (Cremades et al., 2022; Troy et al., 2023). The EPM is a temporally dynamic, valuation-based model of how emotions are generated (level 1, first-order system) and regulated (level 2, second-order system; see Figure 1).

Each stage of the EPM is composed of four components reflecting the *world* (internal or external stimulus to be appraised; W), *perception* (awareness of the stimulus; P), *valuation* (appraisal of the stimulus relative to salient goals; V), and *action* (prepotent behavioral output on the stimulus; A). The first-order emotion generation system receives input from a stimulus (W)

which is perceived (P) and then valued (V) relative to personal goals which produces an affective reaction (A) composed of loosely coupled activation across systems (cognitive, physiological, and behavioral).

The output of the first-order system—an emotion—is then the subject of the second-order emotion regulation system, which unfolds over three stages: identification, selection, and implementation. During *identification*, emotional experience (W) is perceived (P), valued relative to goals and beliefs regarding emotional experience (V) in order to produce an action (A) of a goal to change emotional experience or not (identification of the goal to regulate). During *selection*, available strategies (e.g., cognitive change strategies) within an individual's repertoire are perceived (P) and valued given current emotional goals and context (V) in order to produce an action (A) of a goal to use valued strategies to change emotional experience. During *implementation*, specific tactics (e.g., within cognitive change: reinterpreting the meaning of a stimulus, creating psychological distance, challenging distorted automatic thoughts) and contextual factors are perceived (P) and those tactics are valued given goals and context which may affect their application (V) in order to produce the action (A) of applying a tactic to change emotional experience. Further, these three stages are all subject to *monitoring dynamics* which determine the ongoing interactions among stages (i.e., switching strategies or tactics and identifying when to end regulatory efforts). This process is posited to unfold continuously over time with each cycle being the subject of subsequent cycles (for further details including model development, see Gross, 2015b, 2015a; Sheppes et al., 2015; Webb, Schweiger Gallo, et al., 2012).

Within the EPM, five “families” of strategies are posited to target specific components of emotion generation (Gross, 2015a; McRae & Gross, 2020; Olderbak et al., 2022; Webb, Miles, et

al., 2012). *Situation selection and modification* act on the world (W) through efforts to choose or change stimuli. Examples include avoiding anxiety-evoking situations, pleasurable activity scheduling in behavioral activation, engaging in efforts to solve a problem which is making an environment unpleasant, or introducing pleasant elements to an unpleasant situation. *Attentional deployment* strategies act on perception (P) through focusing attention in ways consistent with emotional goals. Examples include seeking distractions from a stimulus or shifting focus to a stimulus that is consistent with desired emotional experience. *Cognitive change* strategies act on the valuation (V) component through changing how a stimulus is appraised relative to existing goals. Examples include reducing personal relevance or importance, changing determination of how detrimental a stimulus is to goal pursuit, or challenging negative or distorted beliefs about consequences or meanings of a stimulus. *Response modulation* strategies act on the action (A) component through changing the behavioral components of emotion (e.g., facial expressions or approach/avoidance tendencies). Examples include suppressing unwanted emotional expressions or soothing physical sensations; responses made with a goal of altering the world (e.g., avoidance, persistence, problem-solving) are more accurately grouped within situation selection and modification. Within each family are distinct strategies with subordinate tactics of increasing specificity. For instance, within the family of attentional deployment, a person could select the strategy of distraction and implement a specific tactic of thinking about a vacation they recently took.

While each family of strategies targets a specific component of emotion generation, strategy outcomes can also be considered in terms of their effects on emotional experience (*effectiveness*) and functioning (*adaptiveness*). Strategies may be effective in that they achieve hedonic goals (typically, greater positive emotion and less negative emotion) but maladaptive in

that they impair functioning or contribute to psychiatric symptoms (Aldao & Christensen, 2015; Aldao et al., 2010, 2016; Hu et al., 2014). This exact phenomena (a strategy being effective and maladaptive) is applied in models of psychopathology which attribute symptoms partly to the habitual use of avoidance (a form of situation selection or modification; including experiential avoidance) (Ellard et al., 2010; Fernández-Rodríguez et al., 2023; Hayes et al., 2006; Keane et al., 2006). While extensive research has been conducted on the effectiveness of emotion regulation strategies in clinical and non-clinical populations (Dodd et al., 2019; Dryman & Heimberg, 2018; Okur Güney et al., 2019; Picó-Pérez et al., 2017; Webb, Miles, et al., 2012), it is nonetheless vital to also consider the short and long-term adaptiveness of different strategies (Aldao & Christensen, 2015; Aldao et al., 2010, 2016; Eisma & Stroebe, 2021; Hu et al., 2014; Ludwig et al., 2019; Miles et al., 2013; Sloan et al., 2017; Wilson & Gentzler, 2021). Further, no strategy or tactic will be appropriate at all times for all people; it is important to identify the conditions under which a given strategy may be more or less effective or adaptive (Aldao, 2013).

Each stage of the EPM (generation, identification, selection, implementation, and monitoring) can be moderated by personal or contextual factors including psychiatric symptoms, emotional beliefs, features of the stimulus which evoked the emotional response, the emotion(s) to be regulated, available cognitive resources, and others (Dawel et al., 2021; Fergus et al., 2013; Ford & Gross, 2019; Jaso et al., 2020; Karnaze & Levine, 2020; Kneeland et al., 2016; Markovitch et al., 2017; Matthews et al., 2021; Predatu et al., 2020a, 2020b; Shafir & Sheppes, 2020; Sheppes et al., 2009, 2014, 2015; Southward et al., 2019; Suri et al., 2018; Tamir et al., 2007). The identification of these and other moderators inform the application of transdiagnostic psychosocial interventions which aim to improve emotion regulation (or reduce emotion dysregulation) in clinical populations. Specifically, these moderators have been included in

interventions such as the Unified Protocol (Ellard et al., 2010), Acceptance and Commitment Therapy (ACT, Hayes et al., 2006), Dialectical-Behavior Therapy (DBT, Linehan & Wilks, 2015), and Skills Training in Affective and Interpersonal Regulation Narrative Therapy (STAIR, Hassija & Cloitre, 2015). These and other protocols target constructs including emotional awareness, non-acceptance of emotions, distress tolerance, reliance on experiential or behavioral avoidance, and emotion beliefs in order to improve emotion regulation and reduce distress or impairment from psychiatric symptoms.

Given this body of literature, there is increasing interest in: 1) strategies which are both effective and adaptive and 2) moderators which facilitate effective and adaptive emotion regulation. While several strategies or constructs satisfy one of these objectives, mindfulness is of particular interest by satisfying both. Mindfulness-based strategies (such as acceptance) are effective and adaptive (Aldao et al., 2010; Kohl et al., 2012; Naragon-Gainey et al., 2017; Webb, Miles, et al., 2012) while trait mindfulness and participation in mindfulness-based interventions (such as those listed above; mindfulness-based stress reduction, Kabat-Zinn, 2003; and mindfulness-based cognitive therapy, Teasdale et al., 1995) are associated with improved emotion regulation (Fox et al., 2014; Gu et al., 2015; Hill & Updegraff, 2012; Hölzel et al., 2011; Rodrigues et al., 2017; Teper et al., 2013). A therapeutic approach based on the EPM, Emotion Regulation Therapy (ERT, Mennin & Fresco, 2010), emphasizes the use of mindfulness skills to improve emotion regulation and behavioral outcomes among those with distress-based disorders. However, the study of mindfulness in emotion regulation is complicated by difficulty defining mindfulness and a lack of clarity regarding specific mechanisms of action.

Monitoring and Acceptance Theory

There are numerous definitions of mindfulness, including those held by the lay public (Choi et al., 2021), based in “third-wave” psychotherapies such as ACT (Fletcher & Hayes, 2005; Hayes, 2004; Hayes et al., 2011), used in self-report measures of mindfulness (Baer, 2019; Baer et al., 2006), determined by researcher consensus (Bishop et al., 2006), hypothesized in mechanistic models (Lindsay & Creswell, 2017, 2019), and grounded in Buddhist tradition (Grossman & Van Dam, 2011; Grossman, 2019; Kabat-Zinn, 2003; Van Dam et al., 2018). Notably, this variety of definitions also results in vague, contradictory operationalizations and varieties of usage. Mindfulness has been used to mean a specific experiential practice (i.e., mindfulness meditation), a trait or individual difference (dispositional mindfulness), qualities of attention (i.e., mindful attention) separate from acceptance, attitudes of acceptance separate from mindfulness, and qualities of attention characterized by acceptance (Chambers et al., 2009; Van Dam et al., 2018). This may be because mindfulness is a multifaceted and hierarchical construct (Baer et al., 2006; Siegling & Petrides, 2016; Tran et al., 2013, 2014) which is difficult to define analytically due to its experiential nature (Grossman & Van Dam, 2011; Van Dam et al., 2018).

However, the range of definitions does converge on two qualities of mindfulness which can be summarized in the context of Monitoring and Acceptance Theory (MAT, Lindsay & Creswell, 2017, 2019). MAT posits two components of mindfulness with distinct cognitive and affective characteristics. *Monitoring* refers to present-moment awareness of one’s private, internal experiences (thoughts, emotions, physical sensations) and external environment. This perspective on monitoring is differentiated from typical modes of attention through meta-awareness (i.e., attention towards the process of experience rather than the content of experience) (Bernstein et al., 2015, 2019; Fletcher et al., 2010; Grossman & Van Dam, 2011). Acceptance

refers to a particular orientation towards experience characterized by non-judgement, disidentification, non-evaluation, openness, curiosity, compassion, patience, willingness, and decoupling of hedonic tone from goals (Bernstein et al., 2015, 2019; Grossman & Van Dam, 2011; Hadash et al., 2016; Hayes et al., 2006; Williams & Lynn, 2010). The use of “acceptance” is often a shorthand for this overall orientation, which may be more appropriately labeled *equanimity* (Desbordes et al., 2014).

The MAT further posits distinct outcomes through training in monitoring and equanimity. Training in monitoring is thought to strengthen cognitive control (Chiesa et al., 2011; Gill et al., 2020; Im et al., 2021; Levi & Rosenstreich, 2018) and increase reactivity to positive and negative stimuli, although empirical evidence on reactivity is mixed (Inzlicht et al., 2014; Lindsay & Creswell, 2017; Teper et al., 2013; Villalba et al., 2019). Notably, there is a growing body of research on affect labeling (one component of monitoring) as serving an emotion regulation function (Fitzpatrick et al., 2019; Lieberman et al., 2011; Vine et al., 2019), with mixed results (Ortner, 2015). Training in equanimity is thought to reduce emotional reactivity and improve cognitive performance on tasks with affectively valenced material (e.g., emotional Stroop) (Anicha et al., 2012; Jamieson et al., 2022; Lindsay & Creswell, 2017; Teper & Inzlicht, 2013).

Another prominent model, Mindfulness-to-Meaning Theory (MMT, Garland et al., 2017), posits similar mechanisms in the context of self regulation and adaptation to stress, although it generally uses “mindfulness” as an umbrella term. Specifically, the MMT proposes that flexible cognitive control (through monitoring) interacts with reduced threat appraisals (through equanimity) to create an “upward spiral” through which an individual responds adaptively and finds meaning in life stressors. When a stressor is appraised equanimously (i.e., with acceptance), a person is then able to engage in decentering practices without experiential

avoidance. The MMT suggests the metacognitive awareness prompted by a mindful response promotes attentional broadening, flexible responding to stressors, positive emotion, adaptive secondary appraisals, and eudaimonic well-being. Thus, mindfulness skills promote effective self-regulation in the moment and dispositional mindfulness over time.

While monitoring and equanimity are described as separate in order to facilitate analytical, mechanistic understanding of mindfulness, the process of mindfulness is best understood as the continuous interaction of monitoring and equanimity (Bernstein et al., 2015; Chambers et al., 2009; Garland et al., 2017; Grossman & Van Dam, 2011; Grossman, 2019; Hayes, 2004; Hayes et al., 2006; Hölzel et al., 2011; Lindsay & Creswell, 2017, 2019; Van Dam et al., 2018). However, the ways in which this process affects emotion regulation remain to be enumerated.

Dual-mode model of Mindful Emotion Regulation

Emotion regulation and mindfulness are both dynamic, multicomponent, unfolding processes which are likely to interact in dynamic ways. Specifically, the extant literature suggests two “modes” of how mindfulness may act with emotion regulation. To address these modes, provide a framework for ongoing study of mindful emotion regulation, and identify specific processes and mechanisms by which mindfulness interacts with emotion regulation, we propose the **Dual-mode Model of Mindful Emotion Regulation (D-MER)**. This model posits that mindfulness has two modes of action regarding emotion regulation. One mode of action is as a set of discrete strategies which can be implemented to regulate emotions (**implementation**). The other mode of action is as a set of processes which facilitate effective emotion regulation (**facilitation**). These two modes serve to harmonize MAT with the EPM and are outlined in detail below.

[\(Figure 1\)](#)

Implementation

Within implementation, mindfulness skills are specifically applied in a time-limited manner congruent with hedonic and pragmatic goals governed by the EPM stages. In order to change emotional experience, mindfulness skills act on the same emotion generation targets outlined above with the other families of emotion regulation strategies: situation selection or modification, attentional deployment, cognitive change, and response modulation. Effective study of mindfulness implementation requires identification of how mindfulness-based strategies intervene on emotional experience (the “active ingredients”).

Monitoring

To date, there is no clear theoretical consideration of which family of strategies monitoring may reflect. Given how monitoring involves the direction of attention, it is most likely a member of the attentional deployment family. Attentional deployment strategies impact emotion generation through changing how one attends to (perceives) stimuli. Traditionally, this is considered in terms of distraction (directing attention away from affective content) or selective attention (directing top-down attention toward or away from affective content; sometimes referred to as a concentration strategy). Notably, both of these emphasize directing attention regarding the *content* of experience or external stimulus. For example, someone may distract from viewing an unpleasant scene by focusing their attention on a lower-intensity aspect of the scene (such as the background) or on a mental representation (such as geometric shapes).

However, monitoring can be differentiated from these other forms of attentional deployment by shifting attention from content of experience to the *process* of experiencing. Within monitoring, an individual could be attending to a range of different experiences (e.g.,

open monitoring of emotional experiences, focused attention on one's breath, focused repetition of a mantra); however, in all cases there is an emphasis on the process of experience. For example, an experiential form of monitoring may include concentration on breath (content) and maintenance of that concentration despite mind-wandering (process). A more cognitive example can be a thought of "I'm a failure" becoming "I'm thinking I'm a failure" or "I notice that I'm having the thought that I'm a failure" (Bernstein et al., 2015; Lebois et al., 2015). Monitoring can be differentiated from mind-wandering (undirected attention) both in terms of having specific content to focus on and through returning attention when it drifts to other topics (thus reflecting attention towards the process of experience). This has parallels in the broader study of cognition where attentional systems (involving the dorsal and ventral attention networks, Vossel et al., 2014) are differentiated from cognitive control systems, especially action selection and conflict monitoring (involving the anterior cingulate cortex, Walton et al., 2007). While some forms of monitoring training involve specific breathing practices which may serve as response modulation (physical soothing through deep breathing), in such cases there is still an emphasis on the process of attention and maintaining attention on breathing rather than breathing itself as the mechanism of regulation. Some cognitive forms of monitoring strongly resemble the process of cognitive de-fusion, which involves disidentification from experience and is considered an aspect of equanimity rather than monitoring (Hayes et al., 2006). However, this is also consistent with labeling as a form of monitoring. One differentiating factor is that monitoring emphasizes the process of the thought ("I am thinking X") while de-fusion as a component of equanimity emphasizes that a thought does not necessarily reflect a truth and therefore does not require reaction (see below). As such, the D-MER posits that monitoring reflects a form of attentional

deployment distinct from distraction or concentration due to an emphasis on the process of attention and experience.

Equanimity

Equanimity, or acceptance, has received significantly more attention as to which strategy family it may represent (Wojnarowska et al., 2020). Although some have argued for acceptance as a form of response modulation (opposing suppression-based tactics within response modulation, Hofmann & Asmundson, 2008), others have argued it is a form of cognitive change (Chambers et al., 2009; McRae & Gross, 2020; Webb, Miles, et al., 2012). However, the theoretical basis of equanimity as non-evaluative complicates consideration of it as an emotion regulation strategy. While every other emotion regulation strategy aims to change emotional experience, equanimity suggests that emotional experience does not need to be changed.

Although the goal of equanimity is not to change emotional experience, it does change the cognitive valuations of emotions (e.g., “I’m allowed to feel sad”, “this too shall pass”, “this pain is worth it”). While this reflects a form of cognitive change, other cognitive change strategies aim to change the evaluation of the stimulus itself (e.g., finding a “silver lining”, “It isn’t so bad”, or “this is a symptom of my condition, I’m not in danger”). Changing emotional valuations serves a function of emotion regulation (even if that is not the specific goal) and results in different behavioral outputs than experiencing a stimulus without equanimity (e.g., persisting in a task while valuing it negatively). As such, both cognitive change and response modulation are plausible. However, response modulation refers to strategies which change the components of the emotion itself (e.g., physiological responses, facial expressions) rather than the behaviors motivated by the emotion (e.g., withdrawal) (Gross, 2015a; McRae & Gross, 2020). While this distinction is narrow, it is consistent with categorization of avoidance as

situation modification while expressive suppression is a form of response modulation. Further, changes in motivated behavior are one outcome of effective emotion regulation which can differ based on the tactics used to regulate (Aldao & Christensen, 2015). Equanimity reflects a number of related processes including non-judgement, non-evaluation, cognitive de-fusion, disidentification, compassion, and curiosity (Desbordes et al., 2014; Wojnarowska et al., 2020). The function of these processes is changing an individual's subjective valuation of their experience rather than what the experience means or the behavioral components of the emotional reaction. As such, the D-MER posits that equanimity reflects a cognitive change strategy distinct from cognitive reappraisal due to the emphasis on cognitions regarding *reactions* to a stimulus rather than the stimulus itself.

Mindfulness

Given that monitoring and equanimity represent separate families of emotion regulation strategies (attentional deployment and cognitive change), this suggests that mindfulness should be considered a hybrid strategy. Although the EPM is often discussed (as it was above) as a monoregulatory process (i.e., one strategy at a time), theoretical accounts allow for the incorporation of polyregulation (i.e., serial or concurrent application of multiple strategies, Ford et al., 2019). Further, empirical evidence supports that people often engage in spontaneous polyregulation in laboratory settings and daily life (Dixon-Gordon et al., 2015; Naumann et al., 2016; Opitz et al., 2015; Szasz et al., 2018). Thus, the D-MER posits that mindfulness is a hybrid strategy that blends attentional deployment and cognitive change which is theoretically and empirically consistent with the EPM.

Summary of implementation

As outlined above, the D-MER makes three proposals regarding mindfulness-based emotion regulation strategies:

1. Monitoring is a form of attentional deployment characterized by meta-awareness.
2. Equanimity is a form of cognitive change characterized by altering the valuation of emotional experiences.
3. Mindfulness is a hybrid strategy characterized by both attentional deployment and cognitive change.

Further, it is posited that each of these strategies can be effectively implemented to meet hedonic goals. The effectiveness and adaptiveness of equanimity-based strategies (i.e., acceptance) is supported by meta-analysis (Aldao et al., 2010; Kohl et al., 2012; Naragon-Gainey et al., 2017; Webb, Miles, et al., 2012). Individual studies support that mindfulness and monitoring are effective emotion regulation strategies (Keng et al., 2013; Kober et al., 2019; Rosenbaum et al., 2020; Uusberg et al., 2016; Zhang et al., 2019); however, no meta-analysis of such studies has been conducted.

While the discussion above emphasizes the theoretical basis for given classifications, empirical evidence can also inform how these strategies are considered. Strategies within the same family may show related regulatory characteristics. For example, cognitive change strategies are less frequently selected at higher levels of arousal compared to distraction strategies (Shafir & Sheppes, 2020; Sheppes & Levin, 2013) and similar effects can be explored with equanimity and monitoring strategies. Strategies within the same family (e.g., equanimity and cognitive change) are likely to unfold over similar periods of time. Methods with high temporal resolution (e.g., electroencephalography) may be able to identify regulatory strategies

or tactics with similar timeframes. Strategies within the same family may recruit overlapping neural circuits; however, they are unlikely to recruit the exact same systems given different proposed targets of attention or cognitive change. Similar techniques can also be leveraged in order to evaluate mechanisms of facilitation.

Facilitation

Within facilitation, mindfulness skills are considered temporally dynamic qualities of attention or attitudes which moderate the components of the EPM. Through consistent practice, these qualities may become more habitual and exert more consistent effects on emotion and emotion regulation. Thus, moderation may be observed during mindful states (such as during meditation), during periods of greater mindfulness (when a person is engaging in more monitoring and equanimity), following training in mindfulness (such as a mindfulness-based intervention or consistent personal practice), or due to dispositional mindfulness (a person who is more mindful on average based on trait measures). While the effects of mindfulness on emotion regulation can be widespread, it is possible to specify facilitation of particular stages of the EPM (generation, identification, selection, implementation, monitoring) as outcomes of mindfulness. The D-MER posits a series of hypotheses regarding how mindfulness components facilitate each component of the valuation cycle in each stage of the EPM, see Table 1. Themes in these hypotheses and mechanisms based in the Research Domain Criteria (Cuthbert, 2014; Insel et al., 2010) are summarized below (for a thorough review of mindfulness and sustained threat with a similar approach, see Garland et al., 2017; for a thorough review of how mindfulness skills can be used to address distress-related difficulties with emotion regulation, see Mennin & Fresco, 2010; Mennin, 2006; Renna et al., 2017, 2020).

Monitoring is strongly associated with cognitive processes of attention, working memory, and cognitive control (Chiesa et al., 2011; Fox et al., 2014; Gill et al., 2020; Im et al., 2021; Levi & Rosenstreich, 2018; Tang et al., 2015; c.f., Vago et al., 2019), including emotional awareness or differentiation (Norman et al., 2019). Thus, monitoring is likely to moderate components of the EPM which rely predominantly on the ability to orient or maintain attention, monitor performance, inhibit prepotent responses, or hold information in working memory. Prior reviews have made similar proposals, suggesting that cognitive control is one mechanism by which mindfulness practice or skills improve emotion regulation (Garland et al., 2017; Gu et al., 2015; Hölzel et al., 2011; Lindsay & Creswell, 2017, 2019; Teper et al., 2013), although specific subconstructs such as goal selection, response selection, or performance monitoring are typically not specified. Thus, we posit that monitoring mostly affects the perception and action components at each stage of the EPM. These components are those which most rely on cognitive processes such as attention, salience detection, goal representation or updating, performance monitoring, response selection and inhibition, and active maintenance and flexible updating of working memory.

While monitoring affects the cognitive processes involved in emotion regulation, equanimity is likely to exert effects predominantly through the RDoC's negative and positive valence systems. There is a growing empirical literature suggesting that equanimity is associated with reduced affective symptoms and reduced reactivity to negative stimuli, including loss and threat (acute, potential, or sustained) (Bernstein et al., 2019; Carpenter et al., 2019; Creswell et al., 2014; Deng et al., 2020; Desbordes et al., 2012; Eddy et al., 2015; Ford et al., 2018; Fox et al., 2014; Garland et al., 2017; Hadash et al., 2016; Kiken & Shook, 2014; Kohl et al., 2012; c.f., Cosme & Wiens, 2015). While findings are mixed, there is also evidence indicating that

mindfulness training impacts anticipation and response to rewards (Garland, Froeliger, et al., 2015; Hafenbrack & Vohs, 2018; Kirk et al., 2015). Additionally, equanimity is likely to reduce maladaptive emotional beliefs (e.g., that emotions are bad, harmful, or uncontrollable) and aversive reactivity which can impede acting on emotion regulation goals or lead to selection of maladaptive strategies (Delgado et al., 2010; Ford & Gross, 2018, 2019; Karnaze & Levine, 2018, 2020; Kneeland et al., 2016; Mennin & Fresco, 2010; Predatu et al., 2020a; Semcho et al., 2023). Thus, we post that equanimity mostly affects the valuation and action components across stages of the EPM; the components which most rely on cognitive appraisals, emotion beliefs, and meta-emotion.

[\(Table 1\)](#)

This summary suggests clear and specific effects of monitoring and equanimity on emotion regulation components. However, the list in Table 1 is not exhaustive (i.e., important constructs and hypotheses may be omitted) and the available literature suggests that the proposed moderation effects are not mutually exclusive with alternatives (such as propositions outlined in Garland et al., 2017; Mennin & Fresco, 2010). While there is clear evidence that monitoring is associated with multiple cognitive processes, there is also theoretical and empirical support for a role of equanimity in cognitive control, including performance monitoring (Inzlicht et al., 2014; Teper & Inzlicht, 2013; Teper et al., 2013). Further, some evidence suggests that monitoring alone (including emotion labeling) may attenuate emotional reactivity (Fitzpatrick et al., 2019; Fletcher et al., 2010; Lieberman et al., 2011; Vine et al., 2019). The evidence reviewed above also does not address how monitoring or equanimity may moderate automatic physiological regulation—as some have observed (Delgado et al., 2010)—which is implicated in emotion generation. As such, careful dismantling studies are necessary in order to identify specific effects

and improve mechanistic understanding of mindfulness processes. Such dismantling is especially relevant at the action component, which has clear contributions from both monitoring and equanimity. The proposed effects are also consistent with the therapeutic use of mindfulness skills from a range of psychosocial interventions (Ellard et al., 2010; Emmelkamp, 2013; Hassija & Cloitre, 2015; Hayes et al., 2006; Linehan & Wilks, 2015; Mennin & Fresco, 2010; Ost, 2008; Southward et al., 2023); the aim of their presentation here is to provide greater mechanistic specificity in the effects of mindfulness on emotion regulation.

Practice effects

While the D-MER makes specific predictions regarding how implementation and facilitation serve as emotion regulation, it also predicts how these two modes respond to practice. Specifically, the effects of mindfulness practice are posited to differ between the two modes.

Within implementation, we hypothesize that effectiveness for improving emotional experience is stable once basic proficiency with mindfulness skills has been established, while the cognitive demands of using the strategies will decrease with practice. This is separate from someone becoming a more skilled meditator (i.e., having greater focus or being able to meditate for longer periods) and is specifically the ability to use mindfulness skills to regulate emotions. As implementation is a specific, goal-directed application of skills towards hedonic goals, it is likely to always involve explicit effort. However, it is most effortful among those with limited experience (Taylor et al., 2011) and it is unclear how much training is required to reach basic proficiency for implementation. Given that emotion regulation can become “automatic” (i.e., non-conscious use of emotion regulation strategies, Mauss et al., 2007), implementation of mindfulness-based strategies may become more automatic or even implicit with practice and blur the distinction between the implementation and facilitation modes.

Facilitation includes both state processes (i.e., discrete application of monitoring to improve awareness) and trait processes (i.e., habitual engagement in mindfulness and shifts in more stable beliefs about emotions). As such, it is hypothesized to become more effective and less effortful as individuals accumulate mindfulness experience (such as through regular meditation practice). Monitoring and equanimity are not necessarily stable traits but require deliberate practice, cultivation, and application which may become more habitual or automatic over time (Chiesa et al., 2013; Grossman & Van Dam, 2011; Grossman, 2019; Teper et al., 2013). Those trained in mindfulness demonstrate a different factor structure on self-report scales (Gu et al., 2016), especially regarding observing their own experiences. Meditation experts also report qualitatively different experiences of mindfulness characterized by meta-cognition and a perspective that perceptions and valuations are mental events (Poletti et al., 2021). However, the clinical correlates and mechanisms of how mindfulness improves mental health appear to be similar across levels of experience (Burzler et al., 2018; Carpenter et al., 2019). It is currently unclear how long individuals must practice mindfulness before facilitation can be observed, especially given considerations of “dispositional mindfulness”, which is correlated with personality traits of neuroticism, extraversion, and openness to experience and may not reflect specific training (Rau & Williams, 2016).

Discussion

Research implications

The primary research implications from the present review are the predictions of the D-MER, which can inform ongoing research into the effects of mindfulness on emotion regulation. However, the above discussion also emphasizes important considerations for ongoing research independent of the specific hypotheses proposed. The integration of constructs proposed in this

theoretical model is only possible because of the depth of prior work in operationalizing the broad and nuanced constructs of emotion regulation and mindfulness. While these operationalizations are well-represented within their respective areas, greater adoption across research areas may facilitate future research with highly specific, mechanistic hypotheses (see Table 2). Much of the research above that emphasized mindfulness used simplified theories of emotion regulation, while much emotion regulation research employs a broad conceptualization of mindfulness without attending to specific components.

Further, clarification of the implementation and facilitation modes may aid in the selection and creation of measures which address the specific questions of interest within that mode. For example, questions regarding implementation may choose to emphasize instructions to use different strategies to evaluate their effectiveness for different purposes (e.g., in a laboratory instructed emotion regulation paradigm). Such manipulation of what skills people use to regulate their emotions can be used to identify the neural and cognitive processes, strengths, limitations, and contextual moderators of mindfulness-based strategies. In contrast, questions regarding facilitation lend themselves to the use of self-report measures and mindfulness inductions. These allow identification of trait-like relations (using self-report) or state-bound effects (such as performance on an emotion regulation task following a mindfulness induction). Intensive longitudinal designs such as ecological momentary assessment or digital phenotyping also offer flexibility to address questions of facilitation or naturalistic implementation, although researchers should be cautious as the two could be easily conflated. There are also numerous avenues of applied clinical research which the D-MER can inform, including dismantling and treatment mechanism research.

Mechanisms of mindfulness-based interventions are of key interest in the refinement and development of psychosocial interventions. However, the limited specificity with which mindfulness is sometimes studied or minimal dismantling of monitoring and equanimity components limits the conclusions which can be drawn from existing research, including in this review. However, the conceptualization of mindfulness and mechanisms posited by the D-MER may provide greater specificity for ongoing clinical research. Future dismantling studies may also follow examples such as Lindsay and colleagues (Lindsay, Chin, et al., 2018; Lindsay, Young, et al., 2018) to identify specific mindfulness components and mechanisms. Additionally, D-MER predictions regarding practice and experience may be most effectively evaluated in a treatment context where frequent assessment and active control conditions can provide internal validity not available in quasi-experimental studies of experienced and beginner meditators.

Clinical implications

The D-MER propositions are considered to apply regardless of the presence of psychopathology. However, a key application of the model is to improve emotion regulation among those who experience difficulties with it (i.e., those experiencing psychiatric symptoms). Notably, the constructs and mechanisms reviewed here are present in several empirically supported psychosocial treatments (e.g., ERT, ACT, Unified Protocol, DBT, STAIR, Mindfulness-Based Stress Reduction, Mindfulness-Based Cognitive Therapy) as components of the intervention or treatment targets. In particular, ERT was developed with a similar approach based in the affective science literature and EPM (Mennin & Fresco, 2010; Mennin, 2006; Renna et al., 2017, 2020). ERT proposes that distress disorders (including generalized anxiety disorder, depressive disorders, and others on the internalizing spectrum) demonstrate abnormalities across the emotional response process. People with such disorders experience conflicted motivational

systems, including over-valuation of withdrawal-based motivational responses, which in turn result in greater valuation of maladaptive regulatory responses (such as distraction or negative self-referential processes). The use of these responses, which are often based in avoidance of affective stimuli, negatively reinforce maladaptive regulation and increase future valuation of withdrawal-based responses. ERT uses monitoring skills to enhance emotional clarity and meta-cognition and equanimity skills to promote behavioral activation and pursuit of value and goal-based behaviors. This approach is similar to that recommended within MAT (Lindsay & Creswell, 2019; Lindsay, Chin, et al., 2018; Lindsay, Young, et al., 2018; Rahl et al., 2017; Villalba et al., 2019) and the mechanistic hypotheses proposed in the present review. However, the hypotheses and literature presented here can also apply independent of a specific treatment manual.

For clients presenting with difficulties in emotion regulation, the models and moderators here may inform case conceptualization. For example, a client may report being able to choose and apply effective and adaptive strategies for managing intense emotions (i.e., no noted difficulties with selection or implementation) but have difficulty identifying when to apply these strategies (i.e., difficulties with identification). For such a client, it may be useful to emphasize emotional awareness skills to increase identification rate given their reported effectiveness once the desire to regulate is identified. However, another client may demonstrate notable meta-emotions or negative emotion beliefs (“I shouldn’t be so angry/sad/anxious”) which disrupt their ability to engage in effective emotion regulation. For this client, training in equanimity may be more relevant. In both cases, such interventions are existing components of common treatments (e.g., the STAIR manual introduces emotional awareness skills, the Unified Protocol includes emotion psychoeducation within the first few sessions, and ERT uses both of these components

in sequence). Thus, a case conceptualization informed by the D-MER may facilitate the selection and implementation of components from empirically supported treatments consistent with the empirical literature reviewed here.

Explicit consideration of implementation and facilitation modes may also present distinct clinical outcomes. For example, a client may report that they can effectively implement mindfulness-based skills such as mindful breathing to regulate emotions during the week. However, that same client may still discuss their experiences with an attitude inconsistent with equanimity (e.g., commenting that breathing helped them to distract from anxious thoughts). While this can still be a desirable treatment outcome, it can provide an additional point of data about client progress. This can be especially true for clients with prominent experiential avoidance or who are not showing expected habituation or cognitive change during exposures (i.e., their mindfulness skills may be acting as a safety behavior). Further, a client who begins discussing their experiences with increasing equanimity (indicative of possible facilitation) may itself be a desirable treatment outcome even if they report infrequent implementation. Indeed, such a situation is entirely plausible as the modes are posited to be largely independent. Clients may also experience facilitation as having a “mental gap” between habitual or automatic responses and more deliberate ones (Chambers et al., 2009).

In addition to case conceptualization and treatment outcomes, clinicians may find the definitions and models presented here useful for how they approach clients. Many treatments which incorporate mindfulness (such as ACT and DBT) include recommendations for the clinician to have their own mindfulness practice (Kabat-Zinn, 2003), which can be thought of as implementation deepening understanding and promoting facilitation of clinical practice (which can include a clinician’s emotion regulation) by mindfulness. More specifically, the literature

regarding mindfulness presented here may be useful for clinicians as they operationalize and identify the “active ingredients” of mindfulness they seek to convey to clients. While the term equanimity may be off-putting and not as useful for clinical practice as “acceptance”, the ideas it represents may assist clinicians in communicating mindful acceptance. In particular, clients may easily adopt an attitude of resignation (e.g., “I accept that this situation is miserable”) rather than one which is more equanimous (e.g., “I feel sad that the situation didn’t turn out how I wanted” or “It isn’t how I wanted things to go, but I kept with it”). This attitude of equanimity could also be modeled through clinician reflections, as long as it is done carefully to not invalidate client experiences.

The D-MER and evidence presented in this review may be useful resources for clinicians as much as for researchers, but should be used within the limits of the empirical support provided (Table 1, “Theoretical or empirical support” column) rather than the hypotheses generated. Clinicians interested in using the principles described here may consult the interventions discussed (Emotion Regulation Therapy, Unified Protocol, Dialectical Behavior Therapy, Acceptance and Commitment Therapy, Skills Training in Affective and Interpersonal Regulation Narrative Therapy, Mindfulness-Based Stress Reduction, Mindfulness-Based Cognitive Therapy) for further details and clinical applications.

Summary

The present review proposes a novel model, the Dual-mode model of Mindful Emotion Regulation (D-MER) which integrates prominent models of emotion regulation (the Extended Process Model of Emotion Regulation) and mindfulness (Monitoring and Acceptance Theory and Mindfulness-to-Meaning Theory). The D-MER posits that mindfulness has two coexisting modes of action on emotion regulation which are differentially affected by practice:

Implementation (mindfulness as emotion regulation) and facilitation (mindfulness for emotion regulation). Within implementation, mindfulness-based skills of monitoring and equanimity can be effectively used to improve emotional experience through attentional deployment and cognitive change, respectively. Within facilitation, mindfulness skills of monitoring and equanimity can serve to improve emotional experience and regulation through effects on cognitive processes and positive or negative valence systems, respectively. As individuals develop mindfulness skills through spiritual practice or mindfulness-based interventions, facilitation will become more automatic and less effortful. While implementation will not show improvements in effectiveness after basic proficiency is achieved, it will require less cognitive effort.

[\(Table 2\)](#)

Emotion regulation and mindfulness are two of the most complex and prominent constructs in clinical and affective science. However, the study of their interactions has been complicated by a lack of clarity in conceptualization and hypothesized mechanisms of action. Although limited by the specificity of the available literature and analytic understanding of an experiential process (mindfulness), the D-MER provides a clear framework integrating mindfulness and emotion regulation with specific, mechanistic hypotheses which can be used in future clinical and affective research. Improved understanding of mindful emotion regulation can be used to inform the myriad of interventions which use these constructs and their components to improve psychological health and well-being.

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Table 1. Hypothesized facilitation effects on the Extended Process Model of Emotion Regulation

Component	Hypothesized moderation	Theoretical or empirical support
Emotion generation		
Perception — Awareness of stimuli	Monitoring improves the ability to orient, maintain, and switch attention to goal-relevant stimuli.	Mindfulness training enhances proactive and reactive cognitive control (Aguerre et al., 2021; Chang et al., 2017; Li et al., 2018) and meta-analyses support effects for a range of cognitive processes (Chiesa et al., 2011; Gill et al., 2020; Im et al., 2021; Sumantry & Stewart, 2021) with some support for associated neurobiological features (i.e., greater activation in the prefrontal cortex and anterior cingulate cortex, Chiesa & Serretti, 2010).
Valuation — Valuation of current experiences relative to goal	Equanimity decreases the intensity of negative valuation.	For a theoretical review of how equanimity would reduce the frequency, intensity, and duration of negative emotions, see Desbordes et al. (2014); greater equanimity or training in equanimity is associated with reduced negative reactivity on subjective and neurophysiological indices (Deng et al., 2020; Desbordes et al., 2012; Eddy et al., 2015; Ford et al., 2018; Hadash et al., 2016; Lindsay, Young, et al., 2018); those with greater trait mindfulness or following a mindfulness induction generated fewer negative but not positive appraisals (Kiken & Shook, 2014).
Action — Output of valuation	Monitoring and equanimity attenuate automatic responding, which allows errors to be detected and more appropriate responses selected.	Reduced response bias during an approach/avoidance task following meditation or among those with higher trait equanimity (Juneau et al., 2021); reduced affective Stroop conflict following mindfulness training and greater top-down error monitoring (Allen et al., 2012); equanimity mediated relationship between mindfulness and error monitoring (Teper & Inzlicht, 2013); ERT proposes that monitoring training can improve flexibility regarding approach and avoidance goals (Mennin & Fresco, 2010).
Identification		
Perception — Awareness of emotional experience	Monitoring facilitates emotion labeling and identification.	Self-report mindfulness is associated with improved emotional awareness and emotion differentiation among nonclinical young adults (Fogarty et al., 2015; Vujanovic et al., 2010); meditators score higher on self-report measures of emotional clarity (Nielsen & Kaszniak, 2006); meta-analysis supports that mindfulness-based interventions reduce alexithymia (Norman et al., 2019); neuroimaging indicates that participants show greater activity in areas associated with emotional awareness and coupling of such activity with subjective emotion ratings (i.e., ventrolateral prefrontal cortex and subgenual anterior cingulate cortex, Braden et al., 2016); monitoring skills are included in several psychosocial interventions in order to improve emotional awareness (Ellard et al., 2010; Hassija & Cloitre, 2015; Mennin & Fresco, 2010). However, self-compassion (a component of equanimity) is also associated with greater emotion differentiation (Galili-Weinstock et al., 2020). Additionally, “exhaustive” labeling may reduce

Component	Hypothesized moderation	Theoretical or empirical support
Valuation — Valuation of current emotions relative to goals and valuation of regulation compared to current experience	Equanimity decreases negative valuations of emotional experiences and increases the threshold at which the need to regulate is identified.	emotional clarity (Vine et al., 2019). Mindfulness-based treatment is associated with increases in distress tolerance (Lotan et al., 2013); acceptance mediated effects of mindfulness-based treatment effects on distress tolerance (Nila et al., 2016); psychoeducation regarding the role of emotions and training in emotional acceptance are common components in transdiagnostic interventions (Ellard et al., 2010; Hassija & Cloitre, 2015; Mennin & Fresco, 2010); the benefits of acceptance may be specific to acceptance of mental experiences rather than situations (Ford et al., 2018); theoretical models suggest that a more equanimous orientation towards emotions (i.e., emotions are not bad) is likely to reduce the probability of identifying the need to regulate (Ford & Gross, 2019); mindfulness training can change emotional meta-cognitions and improve emotional comprehension (Delgado et al., 2010).
Action — Activation of a goal to regulate	Equanimity decreases the effects of negative beliefs about emotions or their controllability which impede regulation.	Theoretical models suggest that beliefs that emotions are uncontrollable may inhibit acting on desired emotion regulation (Ford & Gross, 2019; Sheppes et al., 2015); participants randomized to a malleable emotion condition engaged in more emotion regulation than those in a fixed condition (Kneeland et al., 2016).
Selection		
Perception — Perception of possible regulation strategies	Monitoring attenuates attentional narrowing and bias which limit the ability to represent effective strategies while expanding working memory capacity.	Distress can narrow attentional breadth and reduce the number of regulatory strategies participants can generate (Yang et al., 2021); however, mindfulness practice can attenuate attentional narrowing during distress to prevent a “downward spiral” of maladaptive responses to distress (Garland et al., 2010, 2017; Garland, Geschwind, et al., 2015); cognitive control literature highlighted earlier may also facilitate error detection and goal monitoring for the perception of possible regulation strategies (see also Teper et al., 2013). Increases in working memory capacity could theoretically extend to the ability to represent additional possible strategies.
Valuation — Valuation of strategies	Equanimity and associated decreases in reactivity increase valuation of contextually adaptive compared to maladaptive strategies.	Following mindfulness-based treatment, there was a greater effect of cardiac activity on emotion regulation choice (Ardi et al., 2021); trait mindfulness is associated with greater use of acceptance-based strategies in people with multiple sclerosis (Duraney et al., 2022); mindfulness-based treatment also increased flexibility in emotion regulation choice (Alkoby et al., 2019); distraction is selected more than reappraisal for high-intensity negative emotion (Matthews et al., 2021; Sheppes et al., 2014), thus decreases in emotional reactivity may increase rates of reappraisal; beliefs that emotions are helpful is associated with greater use of reappraisal (Karnaze &

Component	Hypothesized moderation	Theoretical or empirical support
Action — Activation of goal to use one or more strategies	Equanimity decreases the effects of negative beliefs about emotions or their controllability which impede regulation.	Levine, 2018); theoretical models suggest that beliefs about the utility of different emotions (which is likely moderated by equanimity) can motivate the selection of strategies to meet utilitarian or hedonic goals (Ford & Gross, 2019; Tamir & Ford, 2012; Tamir et al., 2007); mindfulness is negatively associated with disengagement responses (Zimmer-Gembeck et al., 2021) which are considered maladaptive (Semcho et al., 2023).
		The same mechanisms which support acting on identification are likely to impact acting on selection, see above. However, beliefs about specific strategies (e.g., that reappraisal is too difficult) may inhibit action on some strategies but not others.
Implementation		
Perception — Representation of relevant features of the regulation strategy and the world	Monitoring attenuates attentional narrowing and bias which limit the ability to represent effective tactics while also improving working memory in order to operate on tactics and contextual features.	The same mechanisms which support perception of original stimulus and available strategies are likely to impact perception of tactics and contextual factors which inform tactic valuation, see above. Mindfulness is associated with improved working memory (Chiesa et al., 2011; Levi & Rosenstreich, 2018) and partially mediates of the effects of mindfulness on trait positive and negative affect (Jha et al., 2010); labeling increased listing of plans for problem-solving and reappraisal (Vine et al., 2019).
Valuation — Valuation of specific tactics within strategy given current context	Equanimity and associated decreases in reactivity increase valuation of contextually appropriate tactics.	Given the contextual nature of specific tactics, the reduced reactivity due to equanimity is likely to decrease the urgency to begin strategy application (in contrast with aversive reactivity, Semcho et al., 2023) and thus allow for more appropriate tactics to be valued more highly than those which are more readily available but may fail to represent immediate context and goals.
Action — Applying specific tactics to emotional experience	Monitoring increases the effectiveness of most strategies through improvements across cognitive control subconstructs. Equanimity increases the effectiveness of problem-solving (situation modification) and cognitive change strategies.	Emotion regulation effectiveness across several strategies is associated with greater working memory (Kobayashi et al., 2018; Pan et al., 2022), interoceptive awareness (Füstös et al., 2013), and emotion differentiation (Kalokerinos et al., 2019), while reappraisal is less effective under high working memory load (Gan et al., 2017). This supports that monitoring is likely to have indirect effects on implementation through effects on cognitive processes systems (see above). Those with a history of mindfulness-based therapy showed greater effectiveness with reappraisal than controls or those with past cognitive-behavioral therapy (Troy et al., 2013); prior use of self-compassion to regulate emotions improved subsequent effectiveness of cognitive reappraisal (Diedrich et al., 2016).

Component	Hypothesized moderation	Theoretical or empirical support
Monitoring		
Maintenance — Continuing regulation	Monitoring increases goal-maintenance and performance monitoring.	Given above evidence of associations between monitoring and cognitive control, these same factors are likely to improve monitoring of ongoing regulation efforts across maintaining, stopping, and switching.
Stopping — Ceasing regulation		
Switching — Switching to a different regulation tactic	In addition to the effects of monitoring, equanimity decreases self-judgment which inhibits effective switching.	Greater equanimity is theoretically and empirically associated with reduced negative meta-emotions which are likely to impede appropriate switching should a given strategy or tactic prove ineffective (Ford & Gross, 2019; Predatu et al., 2020b). Effects on monitoring on cognitive processes may also lessen the costs of task switching

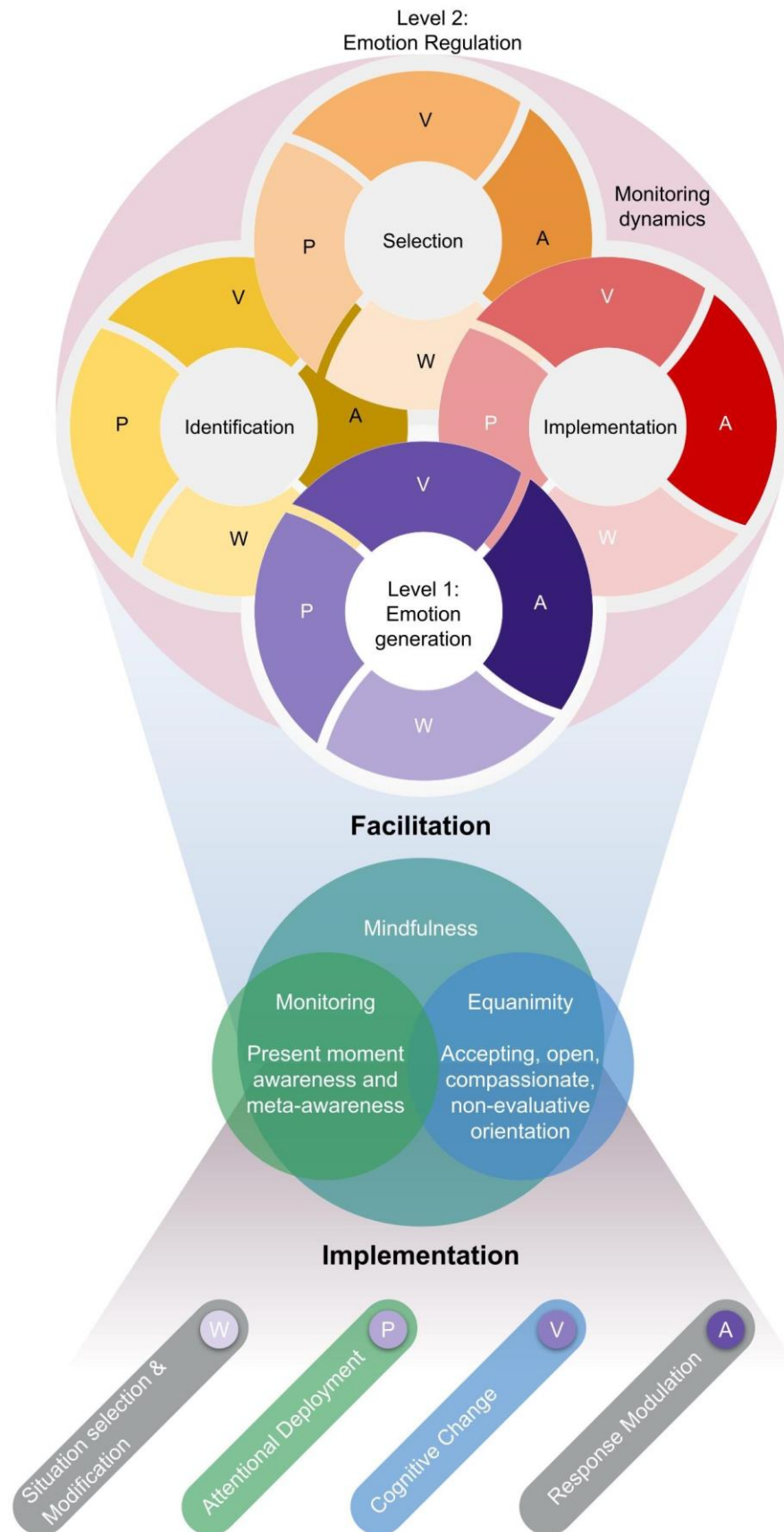
Note. Table format adapted from Sheppes et al. (2015)

Table 2. Summary and hypotheses of the Dual-mode Model of Mindful Emotion Regulation.

Mode	Hypotheses
Implementation — Mindfulness acting as emotion regulation strategies	1. Monitoring modifies how goal-incongruent stimuli are perceived, regulating emotions through shifting of attention towards meta-awareness. 2. Equanimity modifies how emotional responses are valued, regulating them through changing appraisals of emotional responses themselves. 3. Mindfulness is a hybrid strategy composed of monitoring with equanimity. 4. With practice, the use of mindfulness-based strategies will become easier and more automatic; however, it will still require cognitive resources due to effortful implementation.
Facilitation — Mindfulness acting on the emotion regulation process	1. Monitoring skills moderate the perception and action stages of emotion generation and regulation through improvements in attention, working memory, and cognitive control. 2. Equanimity skills moderate the valuation and action stages of emotion generation and regulation through attenuated emotional reactivity and adoption of adaptive emotional beliefs. 3. With practice, the use of mindfulness skills will become more trait-like and require less cognitive effort in order to moderate emotion generation and regulation.

Note. See main text and Table 1 for detailed rationales for hypotheses and moderation hypotheses for each component of each stage of the Extended Process Model of Emotion Regulation.

Figure 1. Dual-mode Model of Mindful Emotion Regulation



Note. W = World, P = Perception, V = Valuation, A = Action. Emotion regulation components are from the Extended Process Model of Emotion Regulation while mindfulness components are from Monitoring and Acceptance Theory. Our model posits that the two mindfulness components act differentially as emotion regulation (Implementation) and on emotion generation and regulation processes (Facilitation). See text, Table 1, and Table 2 for additional details. For more details on the Extended Process Model of Emotion Regulation, see (Gross, 2015b, 2015a; Sheppes et al., 2015; Webb, Schweiger Gallo, et al., 2012). For more details on mindfulness and Monitoring and Acceptance theory, see (Desbordes et al., 2014; Garland et al., 2017; Grossman & Van Dam, 2011; Lindsay & Creswell, 2017, 2019).